(Corrected copy)

MSc Exam 2016

EE9-CS7-2 Game Theory (Model Answers)

1 (i) The mixed strategy for "A": $\frac{1}{2}\times$ row① + $\frac{1}{2}\times$ row② gives new row: $(\frac{1}{2}\times 6+\frac{1}{2}\times 5, \frac{1}{2}\times 6, \frac{1}{2}\times 7) = (5.5, 3, 3.5)$ which dominates row③ : $(6, 4, 4)$

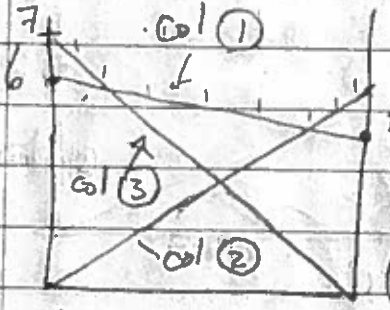
This means "A" will never include row③ in a safety strategy mix

7.6.6 (ii) $\min_i \max_j s_{ij} = \min\{7, 6, 8\} = 6$

$\max_j \min_i s_{ij} = \max\{0, 0, 4\} = 4$

We see max min < min max, i.e. strong minimax condition is not satisfied. But this condition is necessary for existence of a saddle point, So there are no saddle points (pure strategy)

(iii) We remove the dominated 3rd row, to calculate saddle point



Pay-off matrix becomes: $\begin{bmatrix} 6 & 0 & 7 \\ 5 & 6 & 0 \end{bmatrix}$

The safety strategy (for "A") is

$(1-\bar{\lambda})\times$ row① + $\bar{\lambda}$ row② + $0\times$ row③

(because intersection involves rows ① + ②)

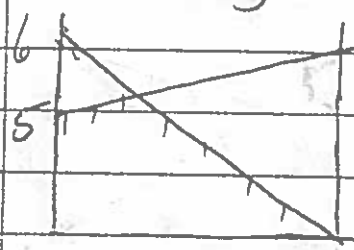
Intersection condition gives

$6(1-\bar{\lambda}) = 5\bar{\lambda} + 6\bar{\lambda}$ $\Rightarrow \bar{\lambda} = 6/7$ ✓

$6(1-\lambda)=\lambda$

Safety strategy for 'A' player is $(\frac{1}{7}, \frac{6}{7}, 0)$ ✓

To calculate safety strategy for 'B' we can ignore column ③ (because Player 'A' will not include it in his safety strategy) Pay-off matrix becomes: $\begin{bmatrix} 6 & 0 \\ 5 & 6 \end{bmatrix}$



Safety strategy for "B" is

$(1-\bar{\mu})\times$ col① + $\mu\times$ col② + $0\times$ col③

Intersection condition is

$(1-\bar{\mu})6 = 5\times(1-\bar{\mu}) + 6\bar{\mu}$

$\Rightarrow \bar{\mu} = 1/7$

$(1-\mu)=6\mu$

Safety strategy for "B" = $(6/7, 1/7, 0)$

Saddle point / Nash equilibrium =

$\left((\frac{1}{7}, \frac{6}{7}, 0), (\frac{6}{7}, \frac{1}{7}, 0)\right)$

(Answers, 2016)

2. Player B plays a feedback strategy depending on whether A chooses F or S. B's strategy space is

$$\begin{pmatrix} p_B(F) \\ p_B(S) \end{pmatrix} \in \left\{\begin{pmatrix} F \\ F \end{pmatrix}\right\} \cup \left\{\begin{pmatrix} F \\ S \end{pmatrix}\right\} \cup \left\{\begin{pmatrix} S \\ F \end{pmatrix}\right\} \cup \left\{\begin{pmatrix} S \\ S \end{pmatrix}\right\}$$

C's strategy space is : $\{F\} \cup \{S\}$

Assume Player A chooses F. Then the pay-offs are

B \ C	F	S
(F,F)	(4,0,4)	(8,4,8)
(F,S)	(4,0,4)	(8,4,8)
(S,F)	(0,8,4)	(4,4,4)
(S,S)	(0,8,4)	(4,4,4)

— (1)

If Player A chooses S, the pay-offs are

B \ C		
→ (F,F)	(0,4,8)	(4,8,8)
→ (F,S)	(8,4,8)	(4,0,4)
→ (S,F)	(0,4,8)	(4,8,8)
→ (S,S)	(8,4,8)	(4,0,4)

— (2)

Player A chooses a mixed strategy

$$0.25 \times (1) + 0.75 \times (2)$$

with average pay-off

B \ C		
(F,F)	(1,3,7)	(5,7,8)
(F,S)	(7,3,7)	(5,1,5)
(S,F)	(0,5,7)	(4,7,7)
(S,S)	(6,5,7)	(2,1,2)

(i) There is a single Nash equilibrium in pure strategies for 'A', 'B'.
B chooses { F if A "=" F
× { F if A "=" S

and C chooses 'F'

The pay-offs are (1, 3, 7) for 'A', 'B', 'C'

(ii) B does not use the information about A's strategy
So B gets no advantage from it.

(Answers, 2016)

3.(i) Assume all players (except 'i') choose mixed strategy
$\bar{\alpha}$: (send), $(1-\bar{\alpha})$: (not send)
But player i chooses mixed strategy
α: (send), $(1-\alpha)$: (not send)
Relevant event space is (for i^{th} player)
{i not send} ∪ {i sends and no-one else sends} ∪ {i sends and at least one other player sends}
pay-off = 2 pay-off = 3 pay-off = 1
Expected pay-off for 'i' is

$$2 \times \text{Prob}\{i \text{ not send}\} + 3 \times \text{Prob}\{i \text{ sends and no-one else sends}\} + 1 \times \text{Prob}\{i \text{ sends and at least one other player sends}\}$$

$$= 2 \times \text{Prob}\{i \text{ not send}\} + 3 \times \text{Prob}\{i \text{ sends}\} \times \text{Prob}\{\text{no one else sends}\} + 1 \times \text{Prob}\{i \text{ sends}\} \times \text{Prob}\{\text{at least one other player sends}\}$$

(by independence of strategies)

$$= 2\times(1-\alpha) + 3\alpha \times (1-\bar{\alpha})^{N-1} + 1 \times \alpha \times \left(1-(1-\bar{\alpha})^{N-1}\right)$$

The expected pay-off must be maximised at $\alpha = \bar{\alpha}$
This is possible (assuming $\bar{\alpha}$ is interior, i.e. $0 < \bar{\alpha} < 1$) only if 'slope' of

$$-2 + [\cancel{\;\;}\; + 3(1-\bar{\alpha})^{N-1} + \left(1-(1-\bar{\alpha})^{N-1}\right]$$

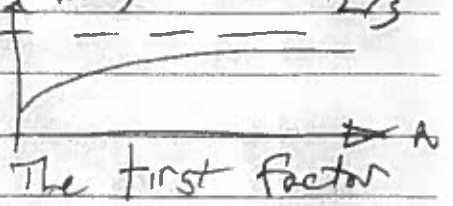
is zero, i.e.

$$1 - 2(1-\bar{\alpha})^{N-1} = 0$$

Hence $\bar{\alpha} = 1 - \left(\frac{1}{2}\right)^{1/N-1}$

So i^{th} player's mixed strategy: $\left(\left(1-\left(\frac{1}{2}\right)^{1/N-1}\right), \left(\frac{1}{2}\right)^{1/N-1}\right)$
(send) (don't send)

(ii) The probability nobody sends is $\left(\frac{1}{2}\right)^{N/N-1} = P(N)$



The probability increases with N.
As N increases the probability 'i' will send decreases, BUT more players might send. The first factor dominates so overall P(N) is increasing.

(Answers, 2016)

4.(i) The pay-offs are

$L^A(P_A, P_B) = P_A(a - P_A - P_B)$ ('A's pay-off)

$L^B(P_A, P_B) = P_B(a - P_A - P_B)$ ('B's pay-off)

For fixed P_B, $L^A(\cdot, P_B)$ is maximized when

$\partial/\partial P_A \; P_A(a - P_A - P_B) = (a - P_A - P_B) - P_A = 0$

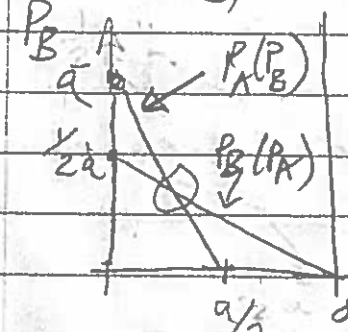
i.e. at $P_A = \frac{1}{2}(a - P_B)$

For fixed P_A, $L^B(P_A, \cdot)$ is maximized at

$P_B = \frac{1}{2}(a - P_A)$ (by symmetry)

Response curves:

$P_A(P_B) = \frac{1}{2}(a - P_B)$, $P_B(P_A) = \frac{1}{2}(a - P_A)$



The diagram shows an intersection at the unique Nash equilibrium.

Nash equilibrium can be got from $\partial/\partial P_A \; L^A(\bar{P}_A, \bar{P}_B) = 0$ and $\partial/\partial P_A \; L^B(\bar{P}_A, \bar{P}_B) = 0$

This gives $a - 2\bar{P}_A - \bar{P}_B = a - 2\bar{P}_B - \bar{P}_A = 0$

$\Rightarrow \bar{P}_A = \bar{P}_B = \frac{1}{3}a$

The pay-off is $\frac{1}{9}a^2$ for both players.

(ii) When company B knows A's price,

Company B maximizes

$L^B(P_A, \cdot) \Rightarrow P_B(P_A) = \frac{1}{2}(a - P_A)$

So Company A maximizes

$P_A \to L^A(P_A, \frac{1}{2}(a - P_A)) = \frac{1}{2}P_A(a - P_A)$

This gives $\frac{1}{2}a - \bar{P}_A \Rightarrow \bar{P}_A = \frac{1}{2}a$

Then $\bar{P}_B = \frac{1}{2}(a - \frac{1}{2}a) = \frac{1}{4}a$.

The pay-offs are $\frac{1}{16}a^2$ (company B) and $\frac{1}{8}a^2$ (company A)

(iii) The optimal pay-offs (for company A) are $\frac{1}{9}a^2 - \frac{1}{3}a^3$ and $\frac{1}{8}a^2 - \frac{1}{3}a^3$ (in the 2 cases)

The maximizing $a = 2/9$ and $1/4$ in the two cases

(Answers 2016)

5 (i) The pay-off is obtained by substituting the state equation into the cost. It is

$$J_\alpha(u_0, v_0; x_0) = (x_0 + u_0 + v_0)^2 + u_0^2 - \alpha v_0^2$$

(ii) The saddle point conditions are

$$\partial_{u_0} J(\bar{u}_0, \bar{v}_0; x_0) = \partial_{v_0} J(\bar{u}_0, \bar{v}_0; x_0) = 0$$

$$\Rightarrow$$

$$2(x_0 + u_0 + v_0) + 2u_0 = 2(x_0 + u_0 + v_0) - 2\alpha v_0 = 0$$

Solving these equations yields

$$\bar{u}_0(\alpha, x_0) = \frac{-\alpha}{(2\alpha - 1)} x_0, \quad \bar{v}_0(\alpha, x_0) = +\frac{1}{2\alpha - 1} x_0.$$

The pay-off is obtained by substituting these formulae into the cost:

$$J_\alpha(\bar{u}_0(\alpha, x_0), \bar{v}_0(\alpha, x_0); x_0) = \frac{(2\alpha + 1)\alpha}{(2\alpha - 1)^2} x_0^2$$

(iii) We see that

$$(\bar{u}_0(\alpha, x_0), \bar{v}_0(\alpha, x_0)) \longrightarrow \left(-\frac{1}{2} x_0, 0\right) \quad \text{as } \alpha \to \infty$$

and

$$J_\alpha(\bar{u}_0(\alpha, x_0), \bar{v}_0(\alpha, x_0); x_0) \longrightarrow \frac{1}{2} x_0^2$$

If we set $v = 0$ (and $\alpha = 0$), the pay-off (w.r.t. u) becomes

$$J_0(u_0, v_0 = 0; x_0) = (x_0 + u_0)^2 + u_0^2$$

The minimizing u_0 satisfies $u_0 = -\frac{1}{2} x_0$

i.e. $u_0^*(x_0) = -\frac{1}{2} x_0$.

We see that $\bar{u}_0(\alpha, x_0) \to u_0^*(x_0)$ as $\alpha \to \infty$

Also the optimal cost is

$$J_0(u_0^*(x_0), v = 0; x_0) = \frac{1}{4} x_0^2 + \frac{1}{4} x_0^2 = \frac{1}{2} x_0^2$$

We see, also, that

$$J_\alpha(\bar{u}_0(\alpha, x_0), \bar{v}_0(\alpha, x_0); x_0 \longrightarrow J_0(u^*(x_0), v = 0; x_0)$$

as $\alpha \to \infty$.